

BE (Comp) / SEM 8 / R19/C - Scheme / SH2023

Upcode: 43043

Duration: - 3 Hours

Marks: 80 Marks

N.B. : (1) Question No 1 is Compulsory.

(2) Attempt any three questions out of the remaining five.

(3) All questions carry equal marks.

(4) Assume suitable data, if required and state it clearly.

- Q.1** Solve any four 20 (4x5)
- Explain hyperlink Analytics?
 - Specify the significance of Social Media KPI
 - Differentiate between static and dynamic social media text.
 - Explain the steps needed to formulate a social media strategy.
 - Explain The benefits of brand building.
- Q.2**
 - Explain the two main categories of search engine analytics. 10
 - What is text analytics, and Explain the steps in text analytics with an example. 10
- Q.3**
 - Explain briefly the seven layers of social media analytics with examples. 10
 - What is a social media-based recommendation system and how does it differ from a traditional recommendation system? 10
- Q.4**
 - What is search engine optimization? What are the different methods to do it? 10
 - What is social media risk? Explain the four steps in social media risk management 10
- Q.5**
 - Explain various issues addressed in privacy policy 10
 - What is Location analytics? Explain its significance in context of social media analytics? 10
- Q.6** Write short notes on any two 20 (2x10)
- Challenges of social media analytics
 - Social Media Action analytics
 - Types of Social media risk
 - Intention analysis in social media.

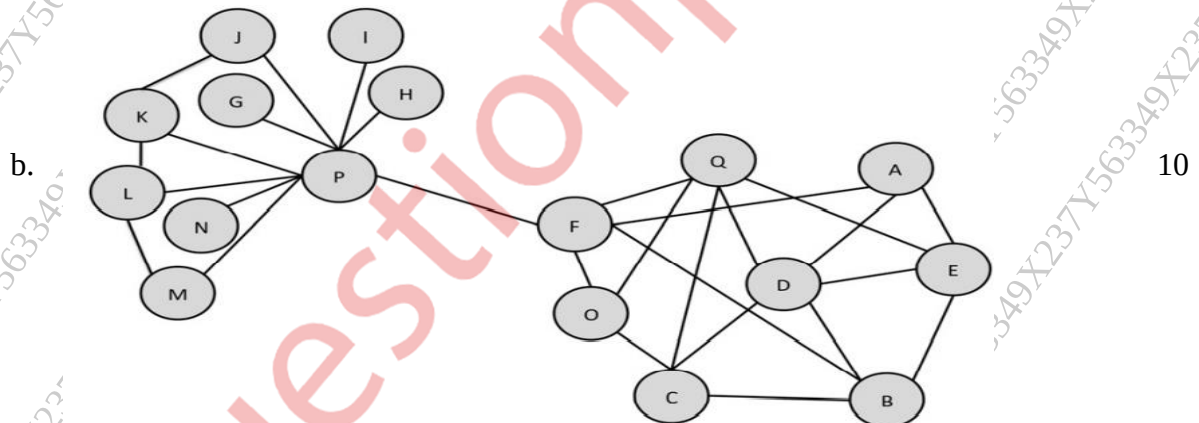
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- Q.1 20
- What is predictive analytics? 5
 - What is text analytics, and why it is useful? 5
 - What is search engine analytics? 5
 - Explain the steps needed to formulate a social media strategy. 5

- Q.2 10
- Differentiate among social media, Web 2.0, and social network sites.
 How degree distribution is plotted for the graph? Show degree distribution of the following graph. 10



- Q.3 10
- Explain Social Media Action Analytics, Common Social Media Actions and Actions Analytics Tools. 10
 - Explain tools of Hyperlink Analytics. 10
- Q.4 10
- List all the location analytics tools and also explain working of every tool. 10
 - What is social media risk? Explain the four steps in social media risk management 10
- Q.5 10
- Discuss various privacy attributes of Social Media Sites. 10
 - What is Location analytics? Explain its significance in context of social media analytics? 10
- Q.6 20
- Write short notes on any two 20
 - Centralization in social media analytics with example.
 - Challenges of social media analytics.
 - Automated, Traditional and Social recommender systems.
 - Social Media Risks Management Framework.

BE Sem VIII (Comp) R-19 C-scheme 19-05-2025

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Marks: 80 Marks

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Q.1 Solve any four

20

- a. Explain Action Analytics with example.
- b. What is the role of Centralization and Tie Strength in social network analysis?
- c. Highlight the differences between Social Media Text Analytics and Hyperlink Analytics.
- d. Explain the challenges faced while performing Social Media Analytics.
- e. Explain how Social Network Visualization enhances the interpretation of large-scale online interactions.

Q.2 a. What is social network structure? List at least two different networks that exist within Instagram. For each one, answer the following: 10

- i. What constitutes a node?
- ii. What constitutes an edge?
- iii. Is it directed?
- iv. Is it weighted? If so, what does the weight indicate?
- v. What is the smallest component in the graph?

b. Explain each of the seven layers in Social Media Analytics and how their integration gives a complete view for business intelligence, with real-life examples 10

Q.3 a. Discuss the role of text analytics in social media analytics. Using a hypothetical social media dataset, explain the process of extracting meaningful insights using text analytic. 10

b. What is a social media-based recommendation system and how can social media-based recommendation systems be used to improve customer service and support, explain with example? 10

Q.4 a. What is search engine optimization? What are the different methods to do it? consider a small, local bakery in Dadar struggled with low online visibility despite having a loyal customer base. Explain how they can improve the visibility using SEO strategies? 10

b. What are social media KPIs? Identify five essential KPIs for an e-commerce brand launching a new product via influencer marketing on Instagram. How do these KPIs support business decision-making? 10

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Q.5 a. Explain the process of managing misinformation risks on social media. Describe the four steps of risk management in the context of a health awareness campaign. 10

b. Describe how public sector agencies can leverage social media analytics during disaster response with examples.

Q.6 a. What is centralization in social network analysis? Illustrate its role using an example from a political campaign. 20

b. Write short note on

i) Privacy concerns in location analytics with examples

ii) Challenges to Social Media Analytics,





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- b) Explain steps for Text Analytics. Explain static and dynamic text analytics. 10
- Q.4** a) What are different social media KPIs. 10
- b) Explain social media action Analytics and Hyperlink Analytics. 10
- Q. 5** a) Elaborate on Social media issues and privacy policies. 10
- b) What is social media Location Analytics and Search Engine analytics 10
- Q.6** Write short notes on **any 4** 20
- a) Applications of Social Media Analytics
 - b) Collaborative Filtering for recommendation
 - c) Difference between social media analytics and Traditional Business Analytics
 - d) Social Media Analytics tools
 - e) Degree Centrality and Closeness centrality
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(3 Hours)

[Total Marks : 80]

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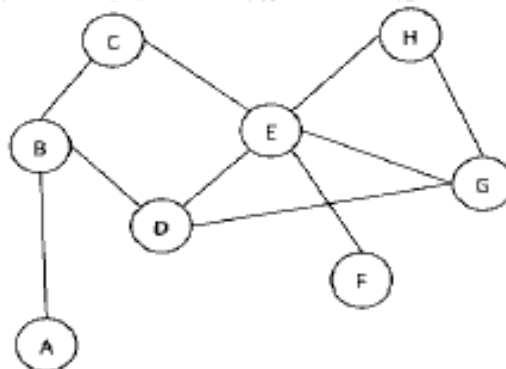
Q.1 Solve any four

20 (4x5)

- a. Define centrality and its types. How is it computed?
- b. Briefly discuss in-links, out-links, and co-links.
- c. What is the purpose of search engine optimization?
- d. Explain the steps needed to formulate a social media strategy.
- e. What are the benefits of social media users who use social media?

Q.2 a. Answer the following questions about this graph.

10



- i. How many nodes are in the network?
 - ii. How many edges are in the network?
 - iii. Is this graph directed or undirected?
 - iv. Create an adjacency list for this graph.
 - v. Create an adjacency matrix for this graph.
 - vi. What is the length of the shortest path from node A to node F?
 - vii. What is the largest clique in this network? How many cliques of that size are there?
 - viii. How many connected components are there in this network?
 - ix. Estimate the density of the graph?
 - x. Are there any hubs in the network? If so, which node (s) and why is it a hub?
- b. Briefly list and define different actions performed by social media users.

10

- Q.3 a. Discuss and differentiate social media texts. 10
- b. Discuss business data-driven location analytics and social media data-driven location analytics? 10
- Q.4 a. Explain the two main categories of search engine analytics. 10
- b. Explain common social media risks-mitigation strategies. 10
- Q.5 a. Briefly explain the seven layers of social media analytics. 10
- b. Explain the ways to measure the success of a company having social media. 10
- Q.6 Write short notes on any two 20 (2x10)
- a. Main challenges to social media analytics.
- b. Sources of Location Data.
- c. Traditional Vs social Recommendation Systems.
- d. Issues with the privacy policies.
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